

Positivity 1.5.C-F

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1 Abelian variety

We first give a useful fact about the divisor on abelian variety.

Theorem 1.1. Let X be an abelian variety over $\mathbb{C}$, which can be written as the quotient V/Λ . Then there is a natural correspondence between the $\mathbb{Z}$ -divisors on X and the Hermite forms on V and there is a equality $(D^n) = n! \cdot \det(h_D)$. Moreover, a divisor D is ample iff the corresponding Hermite form h_D is positive definite.

Proof. Omit. □

The main theorem in this section is the following.

Theorem 1.2. Let X be an abelian variety of dimension n over $\mathbb{C}$, H be an fixed ample integral divisor and D be arbitrary $\mathbb{R}$ -divisor on X . Then D is

1. ample iff $(D^k \cdot H^{n-k}) > 0$ for all $0 \leq k \leq n$.
2. nef iff $(D^k \cdot H^{n-k}) \geq 0$ for all $0 \leq k \leq n$.

Proof. The only if part is trivial by Lazarsfeld example 1.2.5 and 1.4.16. For the if part, we first do some reduction. We can reduce the nef case to the ample case by replacing D by $D + \varepsilon H$ for small ε . Thus we only consider the ample case. Let

$$\mathcal{C} = \{D \in N^1(X)_{\mathbb{R}} \mid (D^k \cdot H^{n-k}) > 0\}$$

We claim that $\mathcal{C} \cap N^1(X)_{\mathbb{Q}} = \text{Amp}(X) \cap N^1(X)_{\mathbb{Q}}$. If this holds, then $\mathcal{C} = \text{Amp}(X)$ since $\mathcal{C}$ and $\text{Amp}(X)$ are both convex cone in $N^1(X)_{\mathbb{R}}$ and $N^1(X)_{\mathbb{Q}} \subseteq N^1(X)_{\mathbb{R}}$ is dense. For the proof of the claim, noting that $\supseteq$ is just the only if part of the theorem which is true, one just need to proof the $\subseteq$ case. We can reduce to the case of $\mathbb{Z}$ -divisors. Now let $P_D(t) = ((D + tH)^n)$. If D is not ample, then h_D is not positively definite. But we know that $D + tH$ is ample for sufficiently large t . Thus by continuity of smallest eigenvalue $\lambda(t)$ of h_D , there must be some $t_0 > 0$ such that $\lambda(t_0) = 0$. i.e. $P_D(t_0) = 0$, which contradicts to the fact that $P_D(t) = \sum_{k=0}^n \binom{n}{k} (D^k \cdot H^{n-k}) t^k > 0$. □

Corollary 1.1. Let X be an abelian variety of dimension n over $\mathbb{C}$ and h resp. $\delta \in N^1(X)$ be an ample resp. nef but not ample class. Then $(\delta^n) = 0$.

Proof. By previous theorem, there exists k such that

$$(\delta^k \cdot h^{n-k}) = 0$$

By Hodge-type inequality 1.1 we know that

$$0 = (\delta^k \cdot h^{n-k})^n \geq (\delta^n)^k \cdot (h^n)^{n-k}$$

But $(h^n) > 0$ by amplitude of h . Thus $(\delta^n) = 0$. □

Lemma 1.1. Let X be an irreducible complete scheme of dimension n over $\mathbb{C}$. Let α, β be nef classes on X . Then for any $0 \leq k \leq n$ we have

$$(\alpha^k \cdot \beta^{n-k})^n \geq (\alpha^n)^k \cdot (\beta^n)^{n-k}$$

Proof. Omit. See Lazarsfeld corollary 1.6.3. □

2 Cone theorem

Theorem 2.1. Let X be a smooth projective variety of dimension n and K_X not nef. Given any divisor D on X , recall that

$$\overline{\text{NE}}(X)_{\geq 0} = \{\alpha \in \overline{\text{NE}}(X) \mid (\alpha \cdot D) \geq 0\}$$

Then we have

1. There are countably many rational curves $C_i \subseteq X$, with

$$0 \leq -(C_i \cdot K_X) \leq n + 1$$

that together with $\overline{\text{NE}}(X)_{K_X \geq 0}$ generate $\overline{\text{NE}}(X)$, i.e.

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0} \cdot [C_i]$$

2. Fix an ample divisor H . Then given any $\varepsilon > 0$, there are only finitely many of these curves $C_1, \dots, C_t$, whose classes lie in the region $(K_X + \varepsilon H)_{\leq 0}$. i.e. we have

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i=1}^t \mathbb{R}_{\geq 0} \cdot [C_i]$$

Proof. Omit, we only give some examples. □

Example 2.1. (Fano variety) Recall that a Fano variety X is a smooth projective variety such that $-K_X$ is ample. Then it satisfies the condition of cone theorem. And we will give some examples of lower dimension Fano varieties.

1. $X = \mathbb{P}^1$. This is the only case of $\dim X = 1$ since $-K_X$ ample iff $\deg(-K_X) > 0$ i.e. $2 - 2g > 0$, which says $g = 0$. In this case, we know by the definition of $\overline{\text{NE}}(X)$ i.e. the closure of effective curves that

$$\overline{\text{NE}}(X) = \mathbb{R}_{\geq 0} \cdot [L]$$

since $N^1(X)_{\mathbb{R}} = \mathbb{R} \cdot [L]$, where $L \subseteq X$ is a line.

This rest three classes are all the Fano varieties of dimension 2, which are called the Del Pezzo surfaces.

2. $X = \mathbb{P}^2$. In this case,

$$\overline{\text{NE}}(X) = \mathbb{R}_{\geq 0} \cdot [H]$$

since $N^1(X)_{\mathbb{R}} = \mathbb{R} \cdot [H]$, where $H \subseteq X$ is a hyperplane.

3. $X = \mathbb{P}^1 \times \mathbb{P}^1$. In this case,

$$\overline{\text{NE}}(X) = \mathbb{R}_{\geq 0} \cdot [\{p\} \times \mathbb{P}^1] \oplus \mathbb{R}_{\geq 0} \cdot [\mathbb{P}^1 \times \{p\}]$$

since the cone is two-dimeisonal with extreme ray $\{p\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{p\}$.

4. $X = \text{Bl}_{p_1 \dots p_k} \mathbb{P}^2$ is the blow-up of $\mathbb{P}^2$ at general points with $1 \leq k \leq 8$. The condition $1 \leq k \leq 8$ will determine the fact that $-K_X = 3H - E_1 - \dots - E_k$ is ample (One just need to check for the case $C = E_i$ and C is the strict transform of some curve of $\mathbb{P}^2$. Thus one can easily understand that the previous condition of blow-up at general points means that not three points are collinear, not six points are on a quadric curve and not eight points are on a cubic curve). So X is indeed a Fano variety. Now we calculate its Mori cone. From the cone theorem we know that the Mori cone is spanned by the extreme rays since $-K_X$ is ample. i.e.

$$\overline{\text{NE}}(X) = \sum_i \mathbb{R}_{\geq 0} \cdot [C_i]$$

where C_i are rational curves. To see that they are (-1) -curves, we use the adjunction formula

$$C \cdot (C + K_X) = 2p_g(C) - 2.$$

The amplitude of $-K_X$ implies that

$$(-K_X \cdot C) \geq 1.$$

Therefore

$$(C^2) \geq 2p_g(C) - 1 \geq -1$$

A contraction theorem from Mori says that in the Fano varieties the curve C with $(C^2) > 0$ is in the interior of the cone and C with $(C^2) = 0$ can be written as the linear combination of (-1) -curves. Thus C_i are all (-1) -curves and this forces $p_g(C) = 0$. If $[C_i] = dH - m_1E_1 - \dots - m_kE_k$, then we have the formula arised from $(C^2) = -1$ and $(C \cdot K_X) = -1$.

$$d^2 - m_1^2 - \dots - m_k^2 = -1$$

$$3d - m_1 - \dots - m_k = 1$$

By Cauchy-Schwarz inequality we know that

$$d^2 + 1 = (m_1^2 + \dots + m_k^2) \geq \frac{1}{k}(m_1 + \dots + m_k)^2 = \frac{1}{k}(3d - 1)^2$$

And the condition $1 \leq k \leq 8$ implies that d has only finite solutions. Thus we only have finite solutions $(d, m_1, \dots, m_k)$. For the rest part, we want to show that every numerical class is actually given by a curve. We use the Riemann-Roch for surface

$$\chi(X, \mathcal{O}_X(C)) = \frac{1}{2}(C \cdot (C - K_X)) + \chi(X, \mathcal{O}_X) = \frac{1}{2}(-1 - (-1)) + 1 = 1.$$

(Here we omit the calculation of $\chi(X, \mathcal{O}_X)$, but one can show by spectral sequence that $H^i(X, \mathcal{O}_X) = H^i(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2})$ and get the result by $\chi(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}) = 1$.) Thus

$$\begin{aligned} 1 &= h^0(X, \mathcal{O}_X(C)) - h^1(X, \mathcal{O}_X(C)) + h^2(X, \mathcal{O}_X(C)) \\ &= h^0(X, \mathcal{O}_X(C)) - h^1(X, \mathcal{O}_X(C)) + h^0(X, \mathcal{O}_X(K_X - C)) \\ &= h^0(X, \mathcal{O}_X(C)) - h^1(X, \mathcal{O}_X(C)) \end{aligned}$$

since $K_X - C$ is not effective by $(K_X - C) \cdot (-K_X) < 0$. Hence $h^0(X, \mathcal{O}_X(C)) \geq 1$ and thus $[C]$ can be represented by an effective curve!

3 Local Structure of Nef Cone

Let X be an irreducible projective scheme of dimension n and $V \subseteq X$ a subvariety. Then

$$\begin{aligned} \varphi_V : N^1(X)_{\mathbb{R}} &\longrightarrow \mathbb{R} \\ \xi &\longmapsto \int_V (\xi^{\dim V}) \end{aligned}$$

determines a real homogeneous polynomial on $N^1(X)_{\mathbb{R}}$ (we explain the meaning of homogeneous polynomial here, if we pick a basis $\{x_i\}$ of $N^1(X)_{\mathbb{R}}$, then by the multilinear of intersection form, φ_V is just a homogeneous polynomial of degree $\dim V$ in variables x_i).

Definition 1. (Null cone) We define the null cone $\mathcal{N}_V \subseteq N^1(X)_{\mathbb{R}}$ of V by

$$\mathcal{N}_V = \{\xi \in N^1(X)_{\mathbb{R}} \mid \varphi_V(\xi) = 0\}.$$

And the nef boundary is defined by $\mathcal{B}_X = \partial \text{Nef}(X)$.

Remark 1. The set $\{\mathcal{N}_V\}_{V \subseteq X}$ is at most countable since $N_k(X)_{\mathbb{Z}} = \frac{Z_k(X)_{\mathbb{Z}}}{\text{Num}_k(X)_{\mathbb{Z}}}$ is a finitely generated abelian group and thus isomorphic to $\mathbb{Z}^{r_k} \oplus T_k$, which is at most countable.

Example 3.1. If $X = \text{Bl}_{p_1 \dots p_k} \mathbb{P}^n$ with p_i general, then under suitable coordinates, we have

$$\mathcal{N}_X = V(x^n - y_1^n - \dots - y_k^n).$$

Proof. Recall that if $\pi : X \rightarrow Y$ is the blow-up of smooth projective variety along smooth subvariety, then

$$\text{Pic}(X) = \pi^* \text{Pic}(Y) \oplus \mathbb{Z} \cdot E, \quad N^1(X) = \pi^* N^1(Y) \oplus \mathbb{Z} \cdot E$$

where E is the exceptional locus. In particular, fixing a hyperplane H on $\mathbb{P}^n$, we know that when $X = \text{Bl}_{p_1 \dots p_k} \mathbb{P}^n$,

$$N^1(X)_{\mathbb{R}} = \mathbb{R} \cdot H \oplus \mathbb{R} \cdot E_1 \oplus \dots \oplus \mathbb{R} \cdot E_k.$$

Assume that $\xi = xH - y_1E_1 - \dots - y_kE_k$, then

$$\mathcal{N}_X = \{\xi \mid (\xi^n) = 0\} = V(x^n - y_1^n - \dots - y_k^n)$$

by the intersection relations

$$(H^n) = 1, \quad (H \cdot E_i) = 0, \quad (E_i^n) = (-1)^{n-1}.$$

□

Theorem 3.1. For any point $\xi \in \mathcal{B}_X$, there exists subvariety $V \subseteq X$ such that $\varphi_V(\xi) = 0$.

Proof. This is just a translation of Nakai-Moishezon criteria for $\mathbb{R}$ -divisors. $\square$

Theorem 3.2. There is an open dense set $\text{CP}(X) \subseteq \mathcal{B}_X$ with the proposition that for every point $\xi \in \text{CP}(X)$, there exists an open neighborhood $U(\xi) \subseteq N^1(X)_{\mathbb{R}}$ together with a subvariety $V \subseteq X$ such that

$$\mathcal{B}_X \cap U(\xi) = \mathcal{K}_V \cap U(\xi)$$

Remark 2. We won't proof the theorem (for the proof, see Lazarsfeld theorem 1.5.29, which uses only Baire's category theorem and some differential geometry). We just give an explanation here. Previously we give a remark that φ_V is a homogeneous polynomial on $N^1(X)_{\mathbb{R}}$, thus the theorem is just saying that locally on $U(\xi)$, the nef boundary is just the hypersurface cut out by φ_V .

4 Product of Higher Genus Curves

Recall that $X = E \times E$ is the product of smooth projective curve of genus $g \geq 2$ and $N^1(X)_{\mathbb{R}} = \mathbb{R} \cdot f_1 \oplus \mathbb{R} \cdot f_2 \oplus \mathbb{R} \cdot \delta$ if E is very general, where $f_1 = [\{p\} \times E]$, $f_2 = [E \times \{p\}]$, $\delta = [\Delta]$. Now let $\delta' = \delta - f_1 - f_2$, then the intersection relations become

$$(\delta' \cdot f_1) = (\delta' \cdot f_2) = 0, \quad (\delta'^2) = -2g.$$

Given $t \in \mathbb{R}_{>0}$, we define the class

$$e_t = t(f_1 + f_2) - \delta'$$

and we want to know that when it is nef. Note that e_t is ample when t is sufficient large since so is $f_1 + f_2$. Thus the interesting thing is the

$$t(E) = \inf\{t > 0 \mid e_t \text{ is nef}\}$$

Definition 2. (Simple branched covering) Let $\pi : X \rightarrow Y$ be a morphism of smooth projective curves, we say that X admits a simple branched covering π if

1. π is finite surjective;
2. Given any point $P \in X$ with $Q = \pi(P)$, we know that there exists a natural map $f^\# : \mathcal{O}_{Y,Q} \rightarrow \mathcal{O}_{X,P}$. Let $\mathfrak{m}_{Y,Q} = (t_Q)$ and $f^\#(t_Q) = u \cdot t_P^{e_P}$, we call e_P the ramification index at P . A simple branched covering should have at most one $e_P = 2$ with other $e_{P'} = 1$ for all $P \neq P' \in \pi^{-1}(Q)$.

The degree of π is defined by $\deg \pi = [K(Y) : K(X)]$.

Theorem 4.1. (Kouvidakis) Assume E admits a simple branched covering $\pi : E \rightarrow \mathbb{P}^1$ of degree $d \leq [\sqrt{g}] + 1$. Then $t(E) = \frac{g}{d-1}$.

Proof. We first give a lemma

Lemma 4.1. Assume that there exists a integral curve $C_0 \subseteq E \times E$ with $[C_0] = e_s = s(f_1 + f_2) - \delta'$ for some $s \leq \sqrt{g}$. Then e_t is nef iff $t \geq \frac{g}{s}$.

Proof. We use Lazarsfeld example 1.4.33, which says that if X is a smooth projective surface and $C_0 \subseteq$ is an irreducible curve with $(C_0^2) \leq 0$, then

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{C_0 \geq 0} + \mathbb{R}_{\geq 0} \cdot [C_0]$$

Note that $(C_0^2) = 2s^2 - 2g \leq 0$ and thus satisfy the condition of example 1.4.33. And by definition we know that e_t is nef iff $(e_t \cdot C) \geq 0$ for all irreducible curves $C \subseteq X$. First we assume that e_t is nef. Then one must have $(e_t \cdot C_0) \geq 0$, which says that $2st - 2g \geq 0$. i.e. $t \geq \frac{g}{s}$. For the converse, we know that $(e_t \cdot C_0) \geq 0$. Hence we just need to check $C \neq C_0$. Since $[C] \in \overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{C_0 \geq 0} + \mathbb{R}_{\geq 0} \cdot [C_0]$, we just need to check that when $t \geq \frac{g}{s}$, every $\alpha \in \overline{\text{NE}}(X)_{C_0 \geq 0}$ satisfies $(e_t \cdot \alpha) \geq 0$.

$$\begin{aligned} (e_t \cdot \alpha) &= (\alpha \cdot e_s) + (\alpha \cdot (t-s)(f_1 + f_2)) \\ &= (\alpha \cdot e_s) + (t-s)(\alpha \cdot (f_1 + f_2)) \geq 0 \end{aligned}$$

Here the first term is non-negative since $\alpha \in \overline{\text{NE}}(X)_{C_0 \geq 0}$, and the second term is non-negative since $f_1 + f_2$ is an ample class and $t \geq \frac{g}{s} \geq \sqrt{g} \geq s$. $\square$

Now return to the theorem. We produce an irreducible curve C_0 having the class $(d-1)(f_1 + f_2) - \delta'$ by $C_0 = \overline{E \times_{\mathbb{P}^1} E - \Delta_E}$. Use the formula $[C_0] + [\Delta_E] = E \times_{\mathbb{P}^1} E$ we know that it has desired class. Here we omit the examination of the irreducibility of C_0 , which uses some further property of simple branched covering. $\square$